

imation-stability for the k -median objective finds a clustering that is very close to the target by using only $O(k)$ one-versus-all queries in the large cluster case, and in addition is faster than the algorithm we present here. In particular, the algorithm for the large clusters case described in [20] (similar to the one we described in Section 5.4 for the k -means objective) can be implemented in $O(|S|^3)$ time, while the one proposed in [68] runs in time $O(|S|k(k + \log |S|))$. [68] use their algorithm to cluster biological datasets in the Pfam [38] and SCOP [56] databases, where the points are proteins and distances are inversely proportional to their sequence similarity. This setting nicely fits the one-versus all queries model because one can use a fast sequence database search program to query a sequence against an entire dataset. The Pfam [38] and SCOP [56] databases are used in biology to observe evolutionary relationships between proteins and to find close relatives of particular proteins. [68] find that for one of these sources they can obtain clusterings that almost exactly match the given classification, and for the other the performance of their algorithm is comparable to that of the best known algorithms using the full distance matrix.

6 Mixture Models

In the previous sections we saw worst case approximation algorithms for various clustering objectives. We also saw examples of how assumptions on the nature of the optimal solution can lead to much better approximation algorithms. In this section we will study a different assumption on how the data is generated in the first place. In the machine learning literature, such assumptions take the form of a probabilistic model for generating a clustering instance. The goal is to cluster correctly (with high probability) an instance generated from the particular model. The most famous and well studied example of this is the Gaussian Mixture Model (GMM)[46]. This will be the main focus of this section. We will illustrate conditions under which datasets arising from such a mixture model can be provably clustered.

Gaussian Mixture Model A univariate Gaussian random variable X , with mean μ and variance σ^2 has the density function $f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{\sigma^2}}$. Similarly, a multivariate Gaussian random variable, $\mathbf{X} \in \Re^n$ has the density function

$$f(\mathbf{x}) = \frac{1}{|\Sigma|^{1/2}(2\pi)^{n/2}}e^{\left(\frac{-1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)}.$$

Here $\boldsymbol{\mu} \in \Re^n$ is called the mean vector and Σ is the $n \times n$ covariance matrix. A special case is the spherical Gaussian for which $\Sigma = \sigma^2 I_n$. Here σ^2 refers to the variance of the Gaussian in any given direction. Consider k n -dimensional Gaussian distributions, $\mathcal{N}(\boldsymbol{\mu}_1, \Sigma_1), \mathcal{N}(\boldsymbol{\mu}_2, \Sigma_2), \dots, \mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k)$. A Gaussian mixture model $\mathcal{M}$ refers to the distribution obtained from a convex combination of such Gaussian. More specifically

$$\mathcal{M} = w_1\mathcal{N}(\boldsymbol{\mu}_1, \Sigma_1) + w_2\mathcal{N}(\boldsymbol{\mu}_2, \Sigma_2) + \dots w_k\mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k).$$

Here $w_i \geq 0$, are called the mixing weights and satisfy $\sum_i w_i = 1$. One can think of a point being generated from $\mathcal{M}$ by first choosing a component Gaussian i , with probability w_i , and then generating a point from the corresponding Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}_i, \Sigma_i)$. Given a data set of m points coming from such a mixture model, a fairly natural question

is to recover the individual components of the mixture model. This is a clustering problem where one wants to cluster the points into k clusters such that the points drawn from the same Gaussian are in a single partition. Notice that unlike in the previous sections, the algorithms designed for mixture models will have probabilistic guarantees. In other words, we would like the clustering algorithm to recover, with high probability, the individual components. Here the probability is over the draw of the m sample points. Another problem one could ask is to approximate the parameters (mean, variance) of each individual component Gaussian. This is known as the parameter estimation problem. It is easy to see that if one could solve the clustering problem approximately optimally, then estimating the parameters of each individual component is also easy. Conversely, after doing parameter estimation one can easily compute the Bayes optimal clustering. To study the clustering problem, one typically assumes separation conditions among the component Gaussians which limit the amount of overlap between them. The most common among them is to assume that the mean vectors of the component Gaussians are far apart. However, there are also scenarios when such separation conditions do not hold (consider two Gaussian which are aligned in an 'X' shape), yet the data can be clustered well. In order to do this, one first does parameter estimation which needs much weaker assumptions. After estimating the parameters, the optimal clustering can be recovered. This is an important reason to study parameter estimation. In the next section we will see examples of some separation conditions and the corresponding clustering algorithms that one can use. Later, we will also look at recent work on parameter estimation under minimal separation conditions.

6.1 Clustering methods

In this section we will look at distance based clustering algorithms for learning a mixture of Gaussians. For simplicity, we will start with the case of k spherical Gaussians in $\mathbb{R}^n$ with means $\{\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \dots, \boldsymbol{\mu}_k\}$ and variance $\Sigma = \sigma^2 I_n$. The algorithms we describe will work under the assumption that the means are far apart. We will call this as the center separation property:

Definition 5 (Center Separation). *A mixture of k identical spherical Gaussians satisfies center separation if $\forall i \neq j$,*

$$\Delta_{i,j} = \|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\| > \beta_{i,j}\sigma$$

The quantity $\beta_{i,j}$ typically depends on k the number of clusters, n the dimensionality of the dataset and w_{min} , the minimum mixing weight. If the spherical Gaussians have different variances σ_i 's, the R.H.S. is replaced by $\beta_{i,j}(\sigma_i + \sigma_j)$. For the case of general Gaussians, σ_i will denote the maximum variance of Gaussian i in any particular direction. One of the earliest results using center separation for clustering is by Dasgupta [32]. We will start with a simple condition that $\beta_{i,j} = C\sqrt{n}$, for some constant $C > 4$ and will also assume that $w_{min} = \Omega(1/k)$. Let's consider a typical point x from a particular Gaussian $\mathcal{N}(\boldsymbol{\mu}_i, \sigma^2 I_n)$. We have $E[\|X - \boldsymbol{\mu}_i\|^2] = E[\sum_{d=1}^n |x_d - \mu_{i,d}|^2] = n\sigma^2$. Now consider two typical points x and y from two different Gaussians $\mathcal{N}(\boldsymbol{\mu}_i, \sigma^2 I_n)$ and $\mathcal{N}(\boldsymbol{\mu}_j, \sigma^2 I_n)$. We have

$$\begin{aligned} E[\|X - Y\|^2] &= E[\|X - \boldsymbol{\mu}_i + \boldsymbol{\mu}_i - \boldsymbol{\mu}_j - (Y - \boldsymbol{\mu}_j)\|^2] \\ &= E[\|X - \boldsymbol{\mu}_i\|^2] + E[\|Y - \boldsymbol{\mu}_j\|^2] + \|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|^2 \\ &\geq 2n\sigma^2 + C^2\sigma^2n \end{aligned}$$

For C large enough (say $C > 4$), we will have that for any two typical points x, y in the same cluster, $\|x - y\|^2 \leq 2\sigma^2 n$. And for any two points in different clusters $\|x - y\|^2 > 18\sigma^2 n$. Using standard concentration bounds we can say that for a sample of size $\text{poly}(n)$, with high probability, all points from a single Gaussian will be closer to each other, than to points from other Gaussians. In this case one could simply create a graph by connecting any two points x, y such that $\|x - y\|^2 \leq 2\sigma^2 n$. It is easy to see that the connected components in this graph will correspond precisely to the individual components of the mixture model. If C is smaller, say 2, one needs a stronger concentration result [10] mentioned below

Lemma 19. *If x, y are picked independently from $N(\mu_i, \sigma^2 I_n)$, then with probability $1 - 1/n^3$, $\|x - y\|^2 \in [2\sigma^2 n(1 - \frac{4\log(n)}{\sqrt{n}}), 2\sigma^2 n(1 + \frac{5\log(n)}{\sqrt{n}})]$.*

Also, as before, one can show that with high probability, for x and y from two different Gaussians, we have $\|x - y\|^2 > 2\sigma^2 n(1 + \frac{4\log(n)}{\sqrt{n}})$. From this it follows that if r is the minimum distance between any two points in the sample, then for any x in Gaussian i and any y in the same Gaussian, we have $\|x - y\|^2 \leq (1 + \frac{4.5\log(n)}{\sqrt{n}})r$. And for a point z in any other Gaussian we have $\|x - z\|^2 > (1 + \frac{4.5\log(n)}{\sqrt{n}})r$. This suggests the following algorithm

Algorithm CLUSTER SPHERICAL GAUSSIANS

1. Let $\mathcal{D}$ be the set of all sample points.
2. **For:** $i = 1$ to k ,
 - (a) Let x_0 and y_0 be such that $\|x_0 - y_0\|^2 = r = \min_{x, y \in \mathcal{D}} \|x - y\|^2$.
 - (b) Let $T = \{y \in \mathcal{D} : \|x_0 - y\|^2 \leq r(1 + \frac{4.5\log n}{\sqrt{n}})\}$.
 - (c) **Output:** T as one of the clusters.

Handling smaller c

For smaller values of C , for example $C < 1$, one cannot in general say that the above strong concentration will hold true. In fact, in order to correctly classify the points, we might need to see points which are much closer to the center of a Gaussian (say at distance less than $\frac{1}{2}\sigma\sqrt{n}$). However, most of the mass of a Gaussian lies in a thin shell around radius of $\sigma\sqrt{n}$. Hence, one might have to see exponentially many samples in order to get a good classification. Dasgupta [32] solves this problem by first projecting the data onto a random $d = O(\log(k)/\epsilon^2)$ dimensional subspace. This has the effect that the center separation property is still preserved up to a factor of $(1 - \epsilon)$. One can now do distance based clustering in this subspace as the number of samples needed will be proportional to 2^d instead of 2^n .

General Gaussians The results of Dasgupta were extended by Arora and Kannan [10] to the case of general Gaussians. They also managed to reduce the required separation between means. They assumed that $\beta_{i,j} = \Omega(\log(n))(R_i + R_j)(\sigma_i + \sigma_j)$. As mentioned before, σ_i denotes the maximum variance of Gaussian i in any direction. R_i denotes the median radius of Gaussian i ⁸. For the case of spherical gaussians, this separation becomes $\Omega(n^{1/4} \log(n)(\sigma_i +$

⁸The radius such that the probability mass within R_i equals $1/2$

σ_j)). Arora and Kannan use isoperimetric inequalities to get strong concentration results for such Gaussians. In particular they show that

Theorem 20. *Given $\beta_{i,j} = \Omega(\log(n)(R_i + R_j))$, there exists a polynomial time algorithm which given at least $m = \frac{n^2 k^2}{\delta^2 w_{\min}^6}$ samples from a mixture of k general Gaussians, solves the clustering problem exactly with probability $(1 - \delta)$.*

Proof Intuition: The first step is to generalize Lemma 19 for the case of general Gaussians. In particular one can show that for x, y are picked at random from a general Gaussian i , with median radius R_i and maximum variance σ_i , we have with high probability

$$2R_i^2 - 18 \log(n) \sigma_i R_i \leq \|x - y\|^2 \leq 2(R_i + 20 \log(n) \sigma_i)^2$$

Similarly, for x, y from different Gaussians i and j , we have with high probability

$$\|x - y\|^2 > 2 \min(R_i^2, R_j^2) + 120 \log(n) (\sigma_i + \sigma_j) (R_i + R_j) + \Omega((\log(n))^2 (\sigma_i^2 + \sigma_j^2)).$$

The above concentration results imply (w.h.p.) that pairwise distances within points from a Gaussian i lie in an interval I_i and distances between Gaussians $I_{i,j}$ lie in the interval $I_{i,j}$. Furthermore, $I_{i,j}$ will be disjoint from the interval corresponding to the Gaussian with smaller value of R_i (Need a figure here). In particular, if one looks at balls of increasing radius around a point from the Gaussian with minimum radius, σ_i , there will be a stage when there exists a gap: i.e., increasing the radius slightly does not include any more points. From the above lemmas, this gap will be roughly $\Omega(\sigma_i)$. Hence, at this stage, we can remove this Gaussian from the data and recurse. This property suggests the following algorithm outline.

Algorithm CLUSTER GENERAL GAUSSIANS

1. Let r be the smallest radius such that $|B(x, r)| > \frac{3}{4} w_{\min} |S|$, for some $x \in \mathcal{D}$.
Here $|S|$ is the size of dataset and $B(x, r)$ denotes the ball of radius r around x .
2. Let σ denote the maximum variance of the Gaussian with the least radius.
Let $\gamma = O(\sqrt{w_{\min} \sigma})$.
3. **While** $\mathcal{D}$ is non-empty,
 - (a) Let s be such that $|B(x, r + s\gamma)| \cap \mathcal{D} = |B(x, r + (s - 1)\gamma)|$.
 - (b) Remove a set T containing all the points from S which are in $B(x, r + s\gamma \log(n))$.
 - (c) **Output:** T as one of the cluster.

One point to mention is that one does not really know beforehand the value of sigma at each iteration. Arora and Kannan [10] get around this by estimating the variance from the data in the ball $B(x, r)$. They then show that this estimate is good enough for the algorithm to work.

6.2 Spectral Algorithms

The algorithms mentioned in the above section need the center separation to grow polynomially with n . This is prohibitively large especially in cases when $k \ll n$. In this section, we look at how spectral techniques can be used to only require the separation to grow with k instead of n .

Algorithmic Intuition In order to remove the dependence on n we would like to project the data such that points from the same Gaussian become much closer while still maintaining the large separation between means. One idea is to do a random projection. However, random projections from n to d dimensions scale each squared distance equally (by factor d/n) and will not give us any advantage. However, consider the case of two spherical gaussians with means μ_1 and μ_2 and variance $\sigma^2 I_n$. Consider projecting all the points to the line joining μ_1 and μ_2 . Now consider any random point x from the first Gaussian. For any unit vector v along the line joining μ_1 and μ_2 we have that $(x - \mu_1) \cdot v$ behaves like a 1-dimensional Gaussian with mean 0 and variance σ^2 . Hence the expected distance of a point x from its mean becomes σ^2 . This means that for any two points in the same Gaussian, the expected squared distance becomes $4\sigma^2$ (as opposed to $2n\sigma^2$). However, the distance between the means remains the same. In fact the above claim is true if we project onto any subspace containing the means. This subspace is exactly characterized by the Singular Value Decomposition (SVD) of the data matrix. This suggests the following algorithm

Algorithm SPECTRAL CLUSTERING

1. Compute the SVD decomposition of the data.
2. Project the data onto the space of top- k right singular vectors.
3. Run a distance based clustering method in this projected space.

Such spectral algorithms were proposed by Vempala and Wang [67] who reduced the separation for spherical Gaussians to $\beta_{i,j} = \Omega(k^{1/4}(\log(n/w_{\min}))^{1/4})$. The case of general Gaussians was studied in [2] who give efficient clustering algorithms for $\beta_{i,j} = (\frac{1}{\sqrt{\min(w_i, w_j)}} + \sqrt{k \log(k \min(2^k, n))})$. [45] give algorithms for general Gaussians for $\beta_{i,j} = \frac{k^{3/2}}{w_{\min}^2}$.

6.3 Parameter Estimation

In the previous sections we looked at the problem of clustering points from a Gaussian Mixture Model. Another important problem is that of estimating the parameters of the component Gaussians. These parameters refer to the mixture weights w_i 's, mean vectors μ_i 's and the covariance matrices Σ_i 's. As mentioned before, if one could do efficiently get a good clustering, then the parameter estimation problem is solved by simply producing empirical estimates from the corresponding clusters. However, there could be scenarios when it is not possible to produce a good clustering. For, ex. consider two one dimensional gaussians with mean 0 and variance σ^2 and $2\sigma^2$. These gaussians have a large overlap and any clustering method will inherently have a large error. On the other hand, let's look at the statistical distance between the two gaussians, i.e., $\int_x |f_1(x) - f_2(x)| dx$. This measures

how much one distribution dominates the other one. It is easy to see that in this case the Gaussian with the higher variance will dominate the other Gaussian almost everywhere. Hence the statistical distance is close to 1. This suggests that information theoretically, one should be able to estimate the parameters of these two mixtures. In this section, we will look at some recent work of Kalai, Moitra, Valiant [44] and Moitra, Valiant [55] in efficient algorithms for estimating the parameters of a Gaussian mixture model. These works make minimal assumption on the nature of the data, namely, that the component gaussians have noticeable statistical distance. Similar results were proven in [23] who also gave algorithms for more general distributions.

The case of two Gaussians:

We will first look at the case of 2 Gaussians in $\mathbb{R}^n$. We will assume that the statistical distance between the gaussians, $D(\mathcal{N}_1, \mathcal{N}_2)$ is noticeable, i.e., $\int_x |f_1(x) - f_2(x)| dx > \alpha$. Kalai et. al [44] show the following theorem

Theorem 21. *Let $\mathcal{M} = w_1 \mathcal{N}_1(\boldsymbol{\mu}_1, \Sigma_1) + w_2 \mathcal{N}_2(\boldsymbol{\mu}_2, \Sigma_2)$ be an isotropic GMM where $D(\mathcal{N}_1, \mathcal{N}_2) > \alpha$. Then, there is an algorithm which outputs $\mathcal{M}' = w'_1 \mathcal{N}'_1(\boldsymbol{\mu}'_1, \Sigma'_1) + w'_2 \mathcal{N}'_2(\boldsymbol{\mu}'_2, \Sigma'_2)$ such that for some permutation $\pi : \{0, 1\} \mapsto \{0, 1\}$ we have,*

$$\begin{aligned} |w_i - w'_{\pi(i)}| &\leq \epsilon \\ \|\boldsymbol{\mu}_i - \boldsymbol{\mu}'_{\pi(i)}\| &\leq \epsilon \\ \|\Sigma_i - \Sigma'_{\pi(i)}\| &\leq \epsilon \end{aligned}$$

The algorithm runs in time $\text{poly}(n, 1/\epsilon, 1/\alpha, 1/w_1, 1/w_2)$.

The condition on the mixture being isotropic is necessary to recover a good additive approximation for the means and the variances since otherwise, one could just scale the data and the estimates will scale proportionately.

Reduction to a one dimensional problem

In order to estimate the mixture parameters, Kalai et. al, reduce the problem to a series of one dimensional learning problems. Consider an arbitrary unit vector v . Suppose we project the data onto the direction of v and let the means of the Gaussians in this projected space be μ'_1 and μ'_2 . Then we have that $\mu_1 = E[x \cdot v] = E[(x - \boldsymbol{\mu}_1) \cdot v] = \boldsymbol{\mu}_1 \cdot v$. Hence, the parameters of the original mean vector are linearly related to the mean in the projected space. Similarly, let's perturb v to get $v' = v + \epsilon(e_i + e_j)$. Here e_i and e_j denote the basis vectors corresponding to coordinates i and j . Let $\sigma_1'^2$ be the variance of the gaussian in the projected space v' . Then writing $\sigma_1'^2 = E[(x \cdot v')^2]$ and expanding, we get that $E[x_i x_j]$ will be linearly related to $\sigma_1'^2$, σ_1^2 and the μ_i 's. Hence, by estimating the parameters correctly over a series of n^2 , one dimensional vectors, one can efficiently recover the original parameters (by solving a system of linear equations).

Solving the one dimensional problem

The one dimensional problem is solved by the method of moments. In particular, define $L_i[\mathcal{M}]$ to be the i th moment for the mixture model $\mathcal{M}$, i.e., $L_i[\mathcal{M}] = E_{x \sim \mathcal{M}}[x^i \mathcal{M}(x)]$. Also define $\hat{L}_i$ to be the empirical i th moment of the data. The algorithm in [44] does a brute force search over the parameter space for the two Gaussians and for a given candidate model $\mathcal{M}'$ computes the first 6 moments. If all the moments are within ϵ of the empirical moments, then

the analysis in [44] shows that the parameters will be $\epsilon^{1/67}$ close to the parameters of the two gaussians. The same claim is also true for learning a mixture of k 1 dimensional gaussians if one goes upto $(4k - 2)$ moments [55]. The search space however will be exponential in k . It is shown in [55] that for learning k one dimensional gaussians, this exponential dependence is unavoidable.

Solving the labeling problem

As noted above, the learning algorithm will solve n^2 , 1-dimensional problems and get parameter estimates for the two gaussians for each 1-dimensional problem. In order to solve for the parameters of the original gaussians, we need to identify for each gaussian, the corresponding n^2 parameters for each of the subproblems. Kalai et. al do this by arguing that if one projects the two gaussians onto a random direction v , with high enough probability, the corresponding parameters for the two projected gaussians will differ by $\text{poly}(\alpha)$. Hence, if one takes small random perturbations of this vector v , the corresponding parameter estimates will be easily distinguishable.

The overall algorithm has the following structure

Algorithm LEARNING A MIXTURE OF TWO GAUSSIANS

1. Choose a random vector v and choose n^2 random perturbations $v_{i,j}$.
2. For each i, j , project the data onto $v_{i,j}$ and solve the one dimensional problem using the method of moments.
3. Solve the labeling problem to identify the n^2 parameter sets corresponding to a single Gaussian.
4. Solve a system of linear equations on this parameter set to obtain the original parameters.

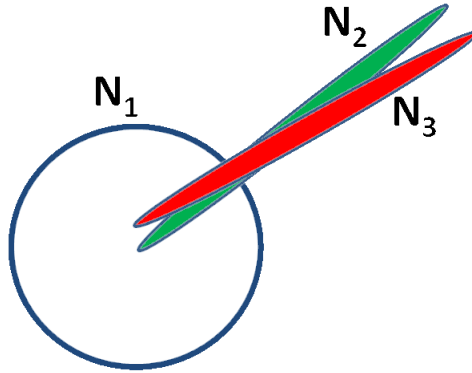


Figure 7: The case of 3 Gaussians.

For the case of more than 2 Gaussians, Moitra and Valiant [55] extend the ideas mentioned above to provide an algorithm for estimating the parameters of a mixture of k Gaussians. For the case of k Gaussians, additional complications arise as it is not true anymore that projecting the k Gaussians to a random 1-dimensional subspace, maintains the statistical distance. For example, consider Figure 7. Here, projecting the data onto a random direction,

will almost surely collapse components 2 and 3. [55] solve this problem by first running a clustering algorithm to separate components 2 and 3 from component 1 and recursively solving the two sub-instances. Once, 2 and 3 have been separated, one can scale the space to ensure that they remain separated over a random projection. The algorithm from [55] has the sample complexity which depends exponentially on k . They also show that this dependence is necessary. One could use the algorithm from [55] to also cluster the points into component Gaussians under minimal assumptions. The sample complexity however, will depend exponentially in k . In contrast, one could algorithms from previous sections to cluster in polynomial time under stronger separation assumptions. The work of [41, 9] removes the exponential dependence on k and designs polynomial time algorithms for clustering data from a GMM under minimal separation assuming only that the mean vectors span a k dimensional subspace. However, their algorithm which is based on Tensor decompositions only works in the case when all the component Gaussians are spherical. It is an open question to get similar result for general Gaussians. There has also been work on clustering points from a mixture of other distributions. [31, 30] gave algorithms for clustering a mixture of heavy tailed distributions. [27] gave algorithms for clustering a mixture of 2 Gaussians assuming only that the two distributions are separated by a hyperplane. The recent work of [49] studies a deterministic separation condition on a set of points and show that any set of points satisfying this condition can be clustered accurately. Using this they easily derive many previously known results for clustering mixture of Gaussians as a corollary.

7 Conclusion

In this chapter we presented a selection of recent work on clustering problems in the computer science community. As is evident, the focus of all these works is on providing efficient algorithms with rigorous guarantees for various clustering problems. In many cases, these guarantees depend on the specific structure and properties of the instance at hand which are captured by stability assumptions and/or distributional assumptions. The study of different stability assumptions also provide insights into the structural properties of real world data and in some cases also lead to practically useful algorithms [68]. As discussed in Section 5.6 different assumptions are suited for different kinds of data and they relate to each other in interesting ways. For instance, perturbation resilience is a much weaker assumption than both ϵ -separability and approximation stability. However, we have algorithms with much stronger guarantees for the latter two. As a practitioner one is often torn between using algorithms with formal guarantees (which are typically slower) vs. fast heuristics like the Lloyd’s method. When dealing with data which may satisfy any of the stability notions proposed in this chapter, a general rule of thumb we suggest is to run the algorithms proposed in this chapter on a smaller random subset of the data and use the solution obtained to initialize fast heuristics like the Lloyd’s method. Current research on clustering algorithms continues to explore more realistic notions of data stability and their implications for practical clustering scenarios.

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